FAHIM AHAMED

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident
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Data Scientist with a strong background in **AI Research**, delivering data-driven solutions across **healthcare**, **cybersecurity**, and **enterprise systems**. Experienced in building **scalable ML pipelines**, conducting **advanced analytics**, and developing **predictive models** that drive measurable outcomes. Skilled in applying **NLP** and **computer vision** to production workflows. Proficient in **Python**, **PyTorch**, **TensorFlow**, **SQL**, and **MLOps**, with a proven ability to turn complex **data challenges** into actionable **business insights**.

EDUCATION

M.S. in Data Science & Engineering - The City College of New York — May 2027
B.S. in Computer Science & Engineering - East West University, Bangladesh — Dec 2023

TECHNICAL SKILLS

Programming Languages: Python, R, Java, C/C++
Data Analysis & Visualization: Tableau, Excel, NumPy, Pandas, Matplotlib, Seaborn
AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK
Database & Big Data: SQL, NoSQL, ChromaDB, Snowflake
MLOps & CI/CD Tools: AWS, MLflow, Git/GitHub, Docker, Streamlit
Others: Data Preprocessing, Feature Engineering, Model Evaluation, A/B Testing, LLMs, RAG, MCP

WORK EXPERIENCE

Data Evangelist Intern - DataCamp — Feb 2026 - May 2026
- Designed and maintained **scalable, reproducible pipelines** to collect, integrate, and validate **multi-source data** using **APIs** and **web scraping**.
- Built and published **data applications** and **analytical workflows**, transforming raw data into **user-facing tools** for exploring key metrics.
- Analyzed data to identify **trends, patterns,** and **key drivers**, producing clear outputs that support **data-driven decision-making.**

Data Analyst - The City College of New York — Oct 2025 - Feb 2026
- Leading the analysis of **extensive institutional data** using **Excel, Tableau,** and **Python**, translating insights into actionable advising and engagement strategies.
- Conducting **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach and support programs.
- Building **analytical reports** and **visual dashboards** that highlight key trends, anomalies, and improvement areas, improving decision-making efficiency across departments.

RESEARCH EXPERIENCE

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis - *Digital Health (SAGE)*
- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline - *IEEE Xplore*
- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques - *Springer Nature*
- **Achieved 97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability - *Springer Nature*
- **Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

ACCEPTED PAPERS (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection - *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

PAPER(S) UNDER REVIEW

Multimodal Learning in Medical Imaging: Methods, Benchmarks, Evaluation, & Clinical Translation - *Elsevier*

- Conducted a structured review of **multimodal learning in medical imaging**, proposing a unified taxonomy while identifying key gaps in **robustness, reproducibility, and benchmark standardization** across healthcare AI systems.

Explainable Token-Fusion Transformer for Early Drowsiness State Recognition in Safety-Critical Systems - *Array Journal*

- Designed **EMTF-ViT**, achieving **99.2% intra-dataset accuracy** and **93%+ cross-dataset transfer**, with Grad-CAM interpretability and real-time low-power deployment for ADAS and safety applications.

Adversarial Energy Latency Sponge Attack on Resource Constraint Devices: A Review - *IEEE IoT Journal.*

- Analyzed and revealed that **sponge attacks** can amplify energy consumption by **up to 30x** and inference latency by **13%**, threatening MIoT and autonomous systems, while proposing **energy-aware defenses** to mitigate these stealth risks.

DATA PROJECTS

NYC HIV/AIDS Trend Analysis & Predictive Modeling — R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret

- Cleaned and engineered **multi-year NYC Open Data** on HIV/AIDS diagnoses, resolving suppressed values, harmonizing demographic fields, and producing analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis of HIV/AIDS trends across NYC**, uncovering persistent geographic and demographic disparities with direct implications for targeted public health interventions.
- **Leveraged tree-based ML models** to support public health planning, with Random Forest ($R^2 = 0.65$) pinpointing high-impact predictors.

Global Population Growth Modeling & Forecasting — R, Tidyverse, ggplot2, leaps, caret

- Engineered a **longitudinal time-series dataset** from World Bank indicators, integrating population, fertility, mortality, migration, and age structure data across **200+ countries**.
- Conducted **multi-decade demographic analysis** (60+ years) by region and income group, identifying structural growth patterns and divergence between developing and developed economies.
- Developed and validated **population growth forecasting models**, achieving **$R^2 = 0.755$**, and applied **rolling-window cross-validation** to assess short- and long-horizon predictive stability.

AI/ML PROJECTS

Clinical Risk Modeling & Unsupervised Structure Analysis — Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

- Curated and standardized **clinical tabular data**, addressing missingness, outliers, and feature scaling to support both supervised prediction and unsupervised structure discovery.
- Performed **diagnostic exploratory analysis** to identify risk-associated variables, revealing substantial class overlaps and highlighting intrinsic limits of linear separability in the dataset.
- Developed a **logistic regression classifier** achieving **~80% test accuracy** and evaluated **K-Means clustering with PCA and t-SNE projections** to assess latent structure and clustering feasibility in medical risk profiling.

Deep Learning Based Wound Classification — Python, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

- **Achieved 98.5% accuracy** and **reduced loss to 0.1061** by refining the CNN architecture and optimizing the training process, significantly boosting model performance.
- **Improved model accuracy by 9%** (from 89.8% to 98.5%) by increasing **architectural complexity** with additional convolutional layers, batch normalization, dropout, and regularization techniques.
- **Enhanced model fairness** and performance by **oversampling underrepresented classes**, effectively addressing challenges posed by significant class imbalance.

Intrusion Detection System — Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

- **Processed and analyzed 2.5+ million network traffic records** through **comprehensive EDA**, uncovering key patterns in anomaly distributions.
- **Reduced memory usage by 47.5%** through down casting numerical data types during preprocessing, enabling more efficient training and analysis of machine learning models.
- **Achieved 97% median accuracy** in detecting network anomalies using machine learning models, enhancing cybersecurity through robust anomaly detection techniques.